CBSE Class 10 Mathematics

Time Allowed: 3 Hours **Maximum Marks: 14**

General Instructions:

1. This question paper contains 11 questions. All questions are compulsory.
2. Questions are grouped into sections A through E.
3. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. If the HCF of two numbers a and b is 12 and their product is 1800, then their LCM is:

(A) (A) 120
(B) (B) 140
(C) (C) 150
(D) (D) 160

2. The decimal expansion of the rational number $\frac{43}{2^4 imes 5^3}$ will terminate after how many places of decimal?

(A) (A) 3
(B) (B) 4
(C) (C) 2
(D) (D) 1

3. If p and q are two prime numbers, then the HCF of p^3q^2 and p^2q is:

(A) (A) pˆ2q
(B) (B) pˆ3qˆ2
(C) (C) pq
(D) (D) pˆ2qˆ2

4. If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$, where x, y are prime numbers, then $LCM(a, b)$ is:

(A) (A) xy
(B) (B) xˆ2yˆ2
(C) (C) xˆ3yˆ3
(D) (D) xˆ4yˆ5

5. The smallest number which is divisible by both 306 and 657 is:

(A) (A) 12338
(B) (B) 22338
(C) (C) 21338
(D) (D) 23338

6. The decimal expansion of the rational number $\frac{43}{2^4 imes 5^3}$ will terminate after how many places of decimal?

(A) (A) 3
(B) (B) 4
(C) (C) 2
(D) (D) 1

Section D: Long Answer (4-5 marks each)

1. If the polynomial $p(x) = x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by $g(x) = x^2 - 2x + k$, the remainder is found to be $x + a$. What is the value of k?

2. When $f(x) = x^3 - 3x^2 + 5x - 3$ is divided by $g(x) = x^2 - 2$, the quotient $q(x)$ is:

3. If $g(x)$ is the divisor of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ and the quotient $q(x) = 2x^2 - 3x + 1$ with $r(x) = 0$, what is the divisor $g(x)$?

4. For a polynomial $p(x)$, if $p(x) = (x^2 - 4)q(x) + (2x + 1)$, what is the remainder when $p(x)$ is divided by $(x - 2)$?

5. A polynomial $p(x)$ of degree 3 is divided by $(x^2 - 1)$ resulting in a quotient $(x - 1)$ and a remainder $(2x + 3)$. Find the polynomial $p(x)$.